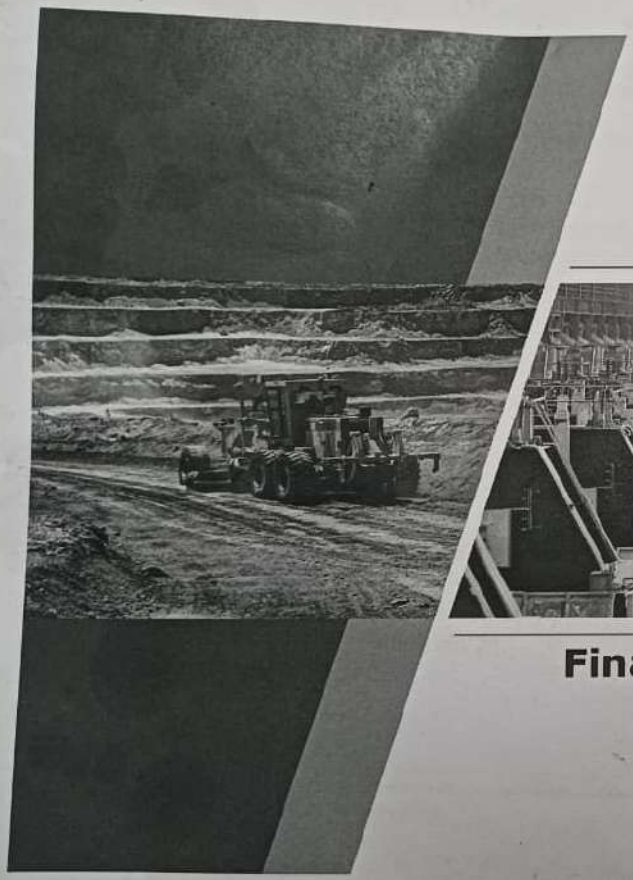
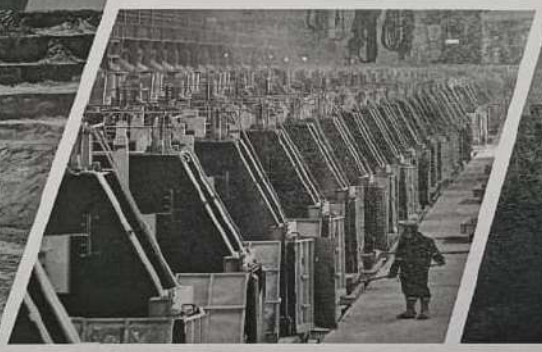


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Final Progress Review Presentation

Name: Ravi Prakash Mishra

Introduction

Name :Ravi Prakash Mishra
Institute: IIT(ISM), Dhanbad.

Dept : Petroleum Engineering.
Anchor/Co-Anchor: Rohit Anil Patil / Praveen Kumar

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Personal & Educational background



UG College: IIT (ISM) , Dhanbad.
Branch: Petroleum Engineering.
Previous Internships: ONGC (Ahmedabad Asset).
Interests: Singing , Athletics.

Professional background and experience

1. Summer Training at ONGC Ahmedabad.
2. Event Management Head at SPE IIT(ISM) Student Chapter.
3. Organizer at Parakram'25 (Sport fest of IIT(ISM), Dhanbad).
4. Student Placement Representative of my batch (2027).
5. Semi-finalist in EY Techackathon'25.

Career Aspiration

>> **Short term:**

- Graduate with good grades and maximize my learning with this internship.
- Learn various other tool for data analysis and Machine learning to use these revolutionary techniques to use in oil and gas operations.

>> **Long term:**

- Lead industry project and implement my learnings to deliver to the industry.
- Contribute towards the growth of country and company with Safety norms and Environmental regulation.

Sensitivity: Internal (C3)

Internship Project – Executive Summary

Name : Ravi Prakash Mishra
Institute: IIT(ISM),Dhanbad

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Internship Start Date	1 June 2026
Internship end Date	31 July 2026

Project Title	1. **Technical project** : Proppant Volume and stage spacing optimization in longitudinally fractured horizontal well of ABH field. 2. **Field visit.**
Anchor	Rohit Anil Patil
Co-anchor/Guide	Praveen Kumar

Project Scope and Deliverables.

1. Trace a diminishing effect Curve for proppant and stage spacing optimization.
2. Maximize the cumulative production while optimizing the stage spacing between different stages and proppant volume used per stage.
3. Get the oil field exposure of Slickline , Eline , Coiled Tubing , SWT.

Project progress.

1. Collect data from various department like frac data for different wells, Production data, End of frac report.
2. Data Preparation and Preprocessing.
3. Plot the graphs and analyze the data.
4. Infer useful results from the graphs and find the optimal points (Proppant volume and stage spacing).

Key Challenges/Learnings

>> Learnings:

- Learn about Hydraulic fracturing technology.
- Factors that affect the fracturing job.
- Learn about how SRT, and minifrac are to be done.
- Learn the importance of optimizing the parameters.
- Learn about the Frac Unit and real time treatment monitoring.

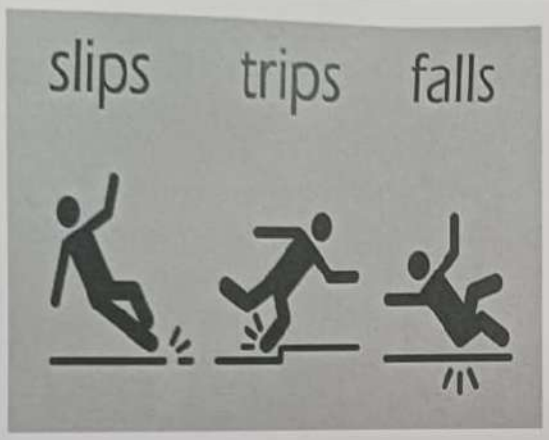
>> Challenges:

- Due to limited number of wells for analysis, we could not find a good curve.
- Manual tracing of datapoints is required.

Sensitivity: Internal (C3)

Safety Moments

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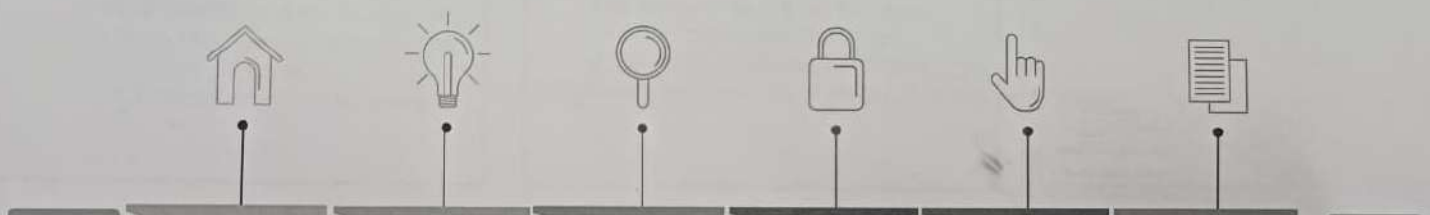




Safety is priority in our job, and it is everyone's responsibility.

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TIMELINE



- *HSE training.*
- *Study FDP of ABH field.*

- *Project Overview*
- *Study of Hydraulic fracturing book from mike smith book and BJ Manual,*
- *Frac Unit Visit.*

- *Developing better understanding of project*
- *Data Collection.*
- *Data Preprocessing.*

- *Research Paper study of similar project in other part of world.*
- *Field visit to Slickline , Eline, Coiled Tubing and SWT.*

- *Data analysis and data visualization.*
- *Interpretation and review.*

- *Presentation work*
- *Final Submission*

Aishwarya Barmer Hill (ABH) Field Study.

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ABH Field Description

1.Discovery : 2006
2.**Reserves : 330 MMSTB(2P),**
P90: 278MMSTB, GIIP: 15.7bcf(2P).
3. Drive Mechanism :Solution Gas Drive.
4.Artificial lift : HRP/ Jet pump.

Reservoir Properties

1.Porosity : 25-35 %.
2.In-situ permeability : 0.1- 4mD.
3.Thickness : 50-250m thick.
4.Temperature: 64 degree Celsius.
5.Reservoir Pressure : 1510 psi @ 820.3mSS



Lithology and location

1.Barmer hill BH formation overlying Fatehgarh formation (Tight oil reservoir).
2.Northern Barmer Basin.
3.Area : 15 sq km.
4. **Closure stress : 3000-4000 psi.**

Fluid Properties

1. Oil API : 30-degree API
2. H2S content : 100 ppm
3. CO2 content : 40-70 mole percent
4. Viscosity : 3 cp
5. WAT : 62.5 degree Celsius.

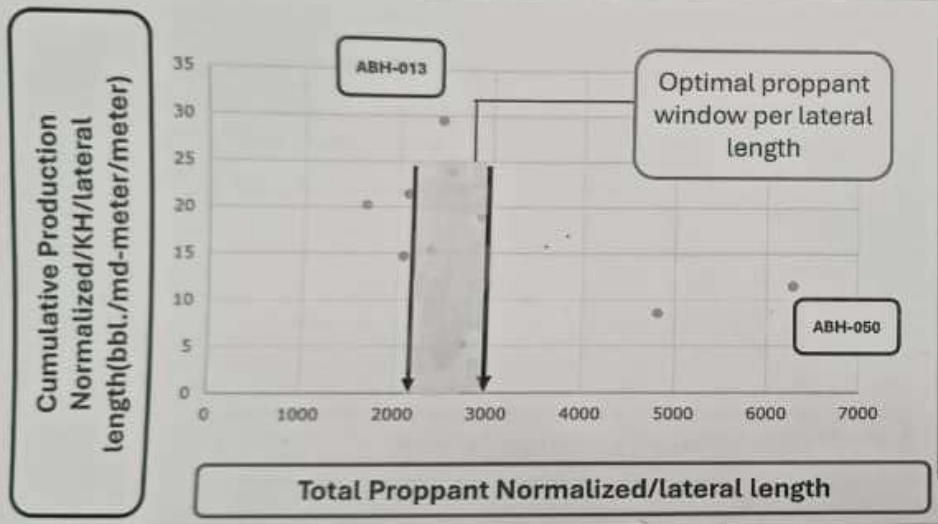


Sensitivity: Internal (C3)

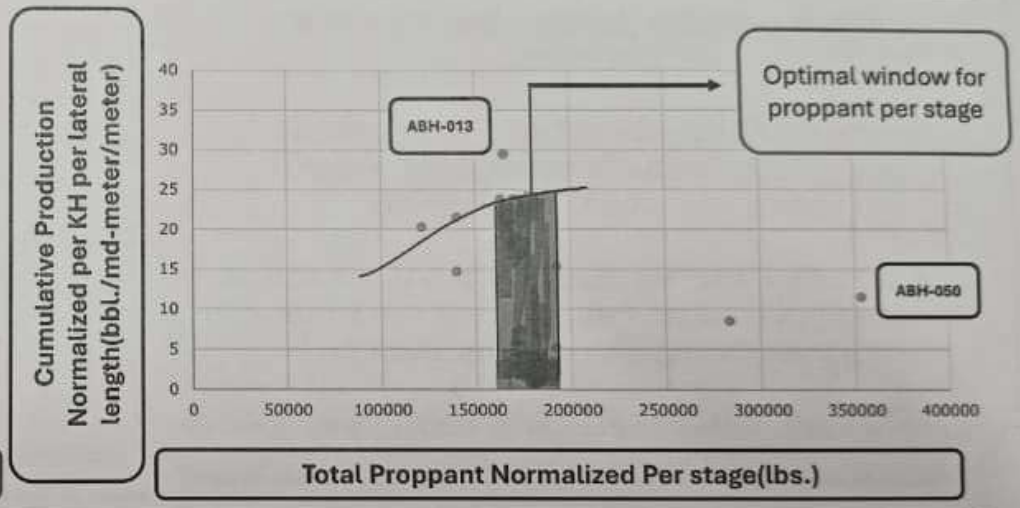
Plots Used for analysis (Longitudinal fractured Horizontal well)

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Cumulative Production per lateral length per KH Vs Total Proppant volume per lateral length



Cumulative production Normalized per KH per lateral length Vs Total Proppant normalized per stage.



Plotted the 12 data points (Cumulative production per stage per KH Vs Total proppant volume per stage) , Result : Using Proppant in range (160K – 180K) gives maximum cumulative production per stage.

Adding more proppant per stage will not lead to any significant increase in oil production.

Abh-050 : 357K proppant per stage : Cumulative production is less than other wells.

Abh-013 : 120K proppant per stage : Cumulative production per stage is more than other wells.

Dimensionless fracture conductivity increase a lot when we add more proppant and may frack the non-contributing zones.

Sensitivity: Internal (C3)

Workflow of this project

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Data Collection.
1) All wells of main block.
2) No GC block wells are to be taken.

Data Points Removal
1) Wells near the fault (ABH-039)
2) New wells with small days production data available.(ABH-066 , ABH-067)

Data Analysis
1) Looking at plots , get a plot of diminishing effect.
2) We get a band of proppant volume and stage spacing.

1 → 2 → 3 → 4 → 5 → 6

Data Preprocessing.

1) Scale the lateral length of all wells to 900 meters.

2) Normalize cumulative production for 1200 days and total proppant accordingly.

3)All wells have different transmittivity, so to normalize the data , divide cumulative production by KH.

Data Plotting and Visualization

1)Cumulative Production per stage per K*H Vs Total proppant per Stage.

2) Cumulative production per Kh per lateral length Vs stage spacing.

2)Plot the above columns in excel.

Inference

1)Optimal proppant volume in longitudinal fractured well is (160k – 180k) lbs.

3) Longitudinal stage spacing: (62-68)meters.

Sensitivity: Internal [C1]

Some other factors on which cumulative production depends

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Type and Concentration of Proppant.	LSP , ISP based on closure stress. Different Proppant >> Different crushing properties. 16/30 , 20/40 , 40/70 >> different prop pack conductivity. Proppant concentration changes with formation response. Higher concentration : resisting closure stress.
Well Completion design.	Variation is perforation strategy and fluid volume used significantly affect the productivity. Wells with similar proppant volume may contribute different in production due to different completion efficiency.
Proppant quality.	Proppant quality decides the nature of fracture. Proppant quality controls fracture conductivity retention, directly influencing well productivity and EUR.
Fracture Geometry	Sometimes fractures grows in non pay zones. Good Fracture geometry doesn't contribute to production due to this reason.
Formation properties.	Every well is different in local geology. Every well have different potential to deliver fluid. Reservoir Heterogeneity.

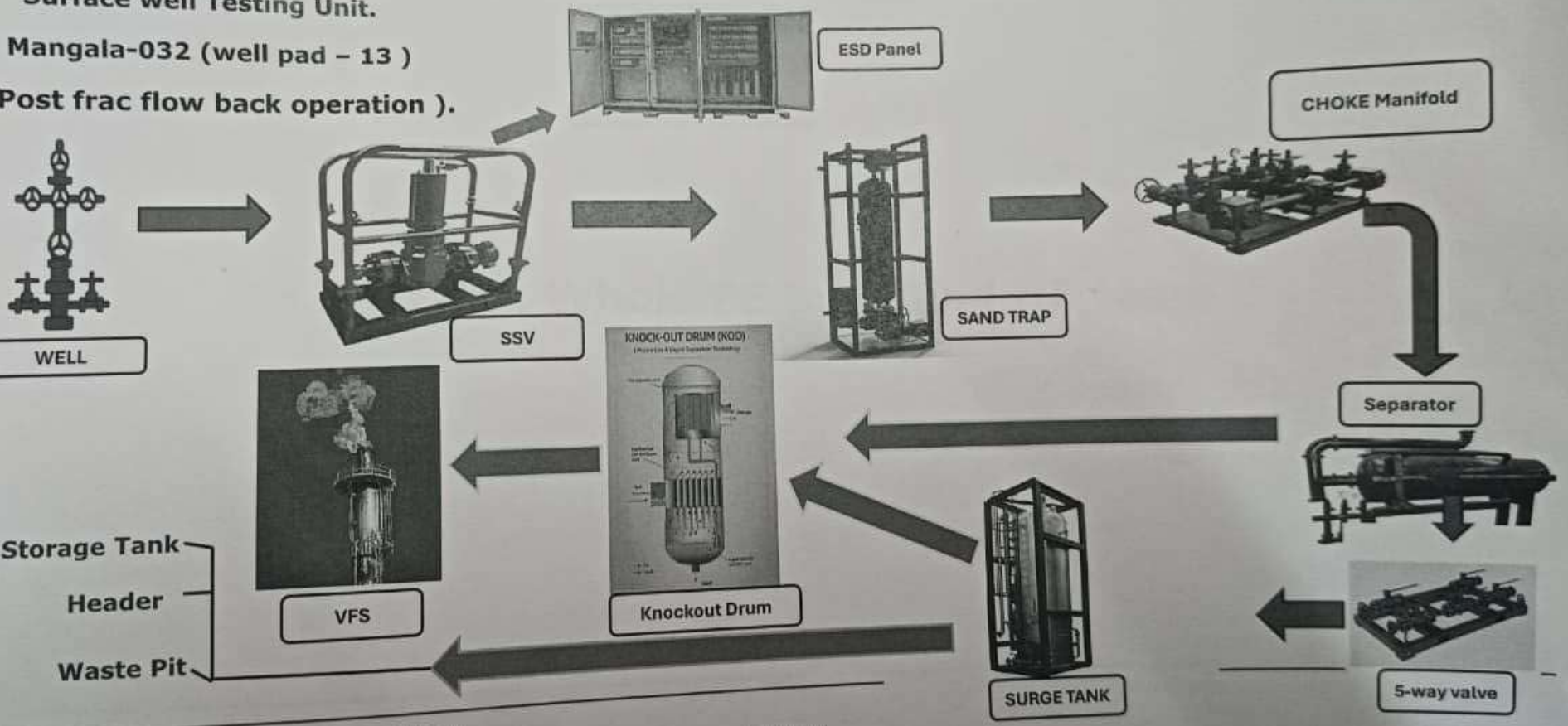
Surface Well Testing (SWT) Unit.

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Surface well Testing Unit.

Mangala-032 (well pad – 13)

(Post frac flow back operation).



Coiled Tubing Unit.

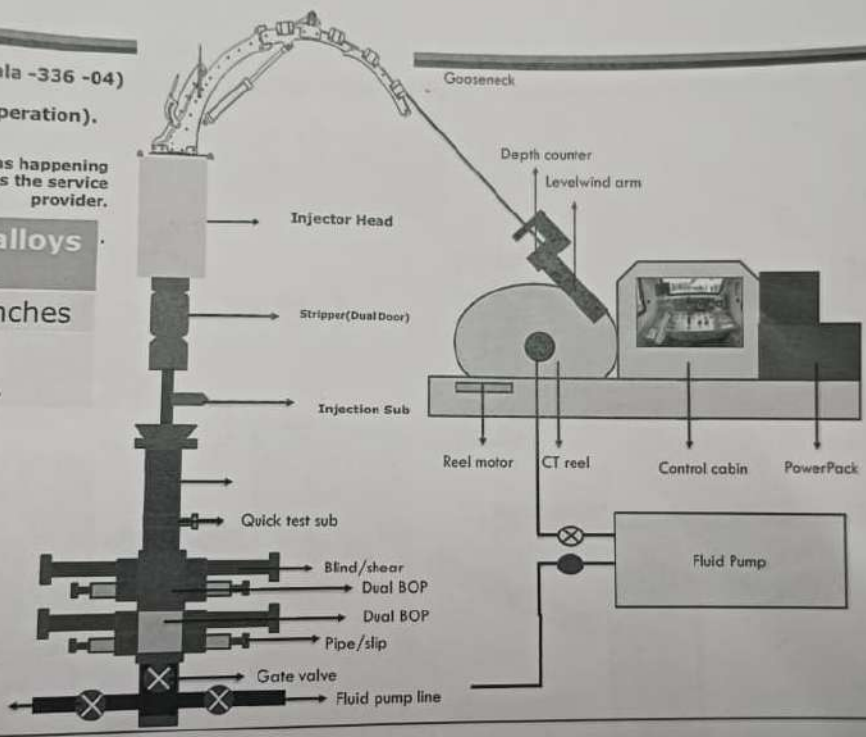
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Coiled Tubing Unit (Magala -336 -04)

(Wellbore cleanout operation).

Wellbore Cleanout Operation was happening in Mangala-336. Baker Huges was the service provider.

Material of tubing	Iron alloys
OD	1.75 inches
ID	1.438 inches.



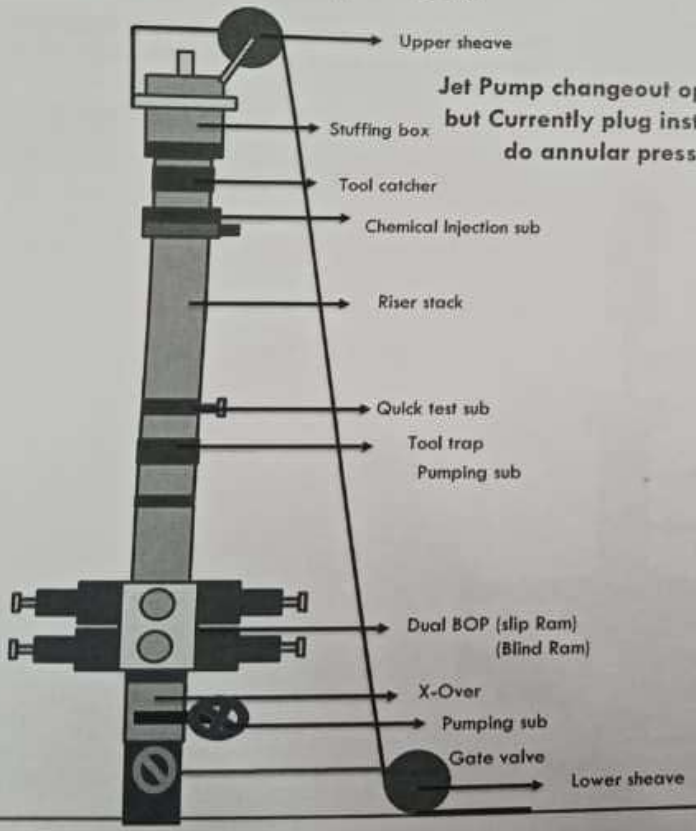
This assembly near the Reel Truck, will have hoses connected with it which provide hydraulic power to all the pressure controlled equipments like injectors, BOP, stripper etc.

Slickline and Eline Unit Field Visit.

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Slickline Unit => Mangala-234-01

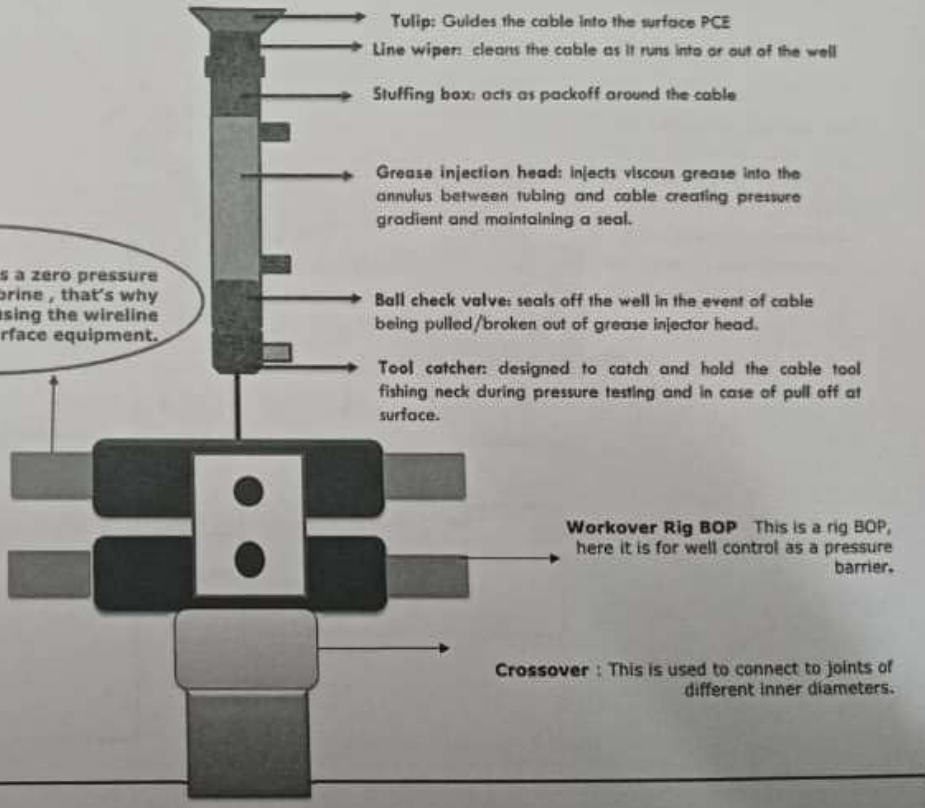
(Jet Pump Changeout operation).



Jet Pump changeout operation was planned but Currently plug installation was done to do annular pressure holdup test.

Eline operation => Mangala -085-01

(RST wireline operation was going on)



Mangala -085 is a zero pressure well filled with brine , that's why they were not using the wireline BOP and other surface equipment.

Tulip: Guides the cable into the surface PCE

Line wiper: cleans the cable as it runs into or out of the well

Stuffing box: acts as packoff around the cable

Grease injection head: injects viscous grease into the annulus between tubing and cable creating pressure gradient and maintaining a seal.

Ball check valve: seals off the well in the event of cable being pulled/broken out of grease injector head.

Tool catcher: designed to catch and hold the cable tool fishing neck during pressure testing and in case of pull off at surface.

Workover Rig BOP This is a rig BOP, here it is for well control as a pressure barrier.

Crossover : This is used to connect to joints of different inner diameters.

Frac Unit : Aish-071-03-110 (Well Pad – 03)

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